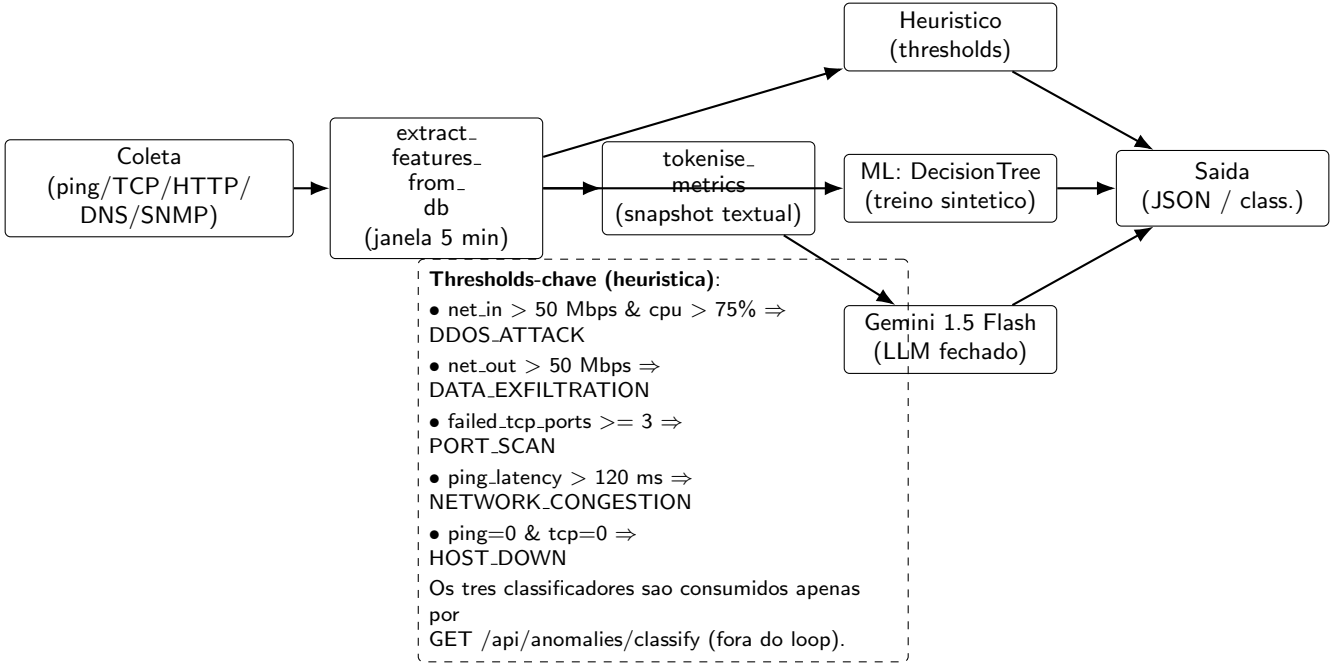

Coleta
(ping/TCP/HTTP/
DNS/SNMP)
extract_
features_
from_
db
(janela 5 min)
Heuristico
(thresholds)
tokenise_
metrics
(snapshot textual)
ML: DecisionTree
(treino sintetico)
Gemini 1.5 Flash
(LLM fechado)
Saida
(JSON / class.)
Thresholds-chave (heuristica):
• net_in > 50 Mbps & cpu > 75% ⇒
DDOS_ATTACK
• net_out > 50 Mbps ⇒
DATA_EXFILTRATION
• failed_tcp_ports >= 3 ⇒
PORT_SCAN
• ping_latency > 120 ms ⇒
NETWORK_CONGESTION
• ping=0 & tcp=0 ⇒
HOST_DOWN
Os tres classificadores sao consumidos apenas
por
GET /api/anomalies/classify (fora do loop).